

TENCENT UNI-REC CHALLENGE 2026

Introduction

Towards Unifying Sequence Modeling and Feature Interaction for Large-scale Recommendation

Recommender systems power both large-scale content platforms (feeds, short videos, etc.) and digital advertising (CTR/CVR prediction, etc.), directly shaping user experience, engagement, and revenue. Operating under massive traffic and strict latency constraints, these systems make billions of real-time decisions daily and underpin a hundred-billion dollar digital advertising market. Over the past two decades, recommendation research has progressed along two major branches: feature interaction models, which focus on modeling high-dimensional multi-field categorical and contextual features, and sequential models, which capture the temporal dynamics of user behavior through embedding-based retrieval systems and Transformer-style ranking models. While both paradigms have achieved significant success, they have largely evolved independently. This separation has created structural bottlenecks in industrial systems: shallow cross-paradigm interaction, inconsistent optimization objectives, limited scalability, and increasing hardware and engineering complexity. As sequence lengths and model scales continue to grow, these fragmented architectures become increasingly inefficient.

In recent years, several works have begun to bridge these two historically separated branches [1–3]. To further accelerate progress in this direction, we introduce the challenge “Towards Unifying Sequence Modeling and Feature Interaction for Large-scale Recommendation”. Participants are encouraged to develop a unified tokenization scheme and a homogeneous, stackable backbone that jointly models sequential user behaviors and non-sequential multi-field features within a single architecture for CVR prediction. Submissions will be ranked by a single AUC of ROC metric. Beyond the leaderboard, we will offer two innovation awards—Unified Block Innovation Award (\$45,000) and Scaling Law Innovation Award (\$45,000)—to recognize outstanding advances in unified architectures and systematic scaling-law exploration, respectively. These awards are independent of leaderboard rank,

and the workshop paper review will emphasize novelty and insight along these two directions rather than focusing solely on AUC.

Dataset

[TAAC2026/data_sample_1000](#) · Datasets at Hugging Face

Columns

Column	Type	Description
<code>item_id</code>	<code>int64</code>	Target item identifier.
<code>item_feature</code>	<code>array[struct]</code>	Array of target item feature dicts. Each element has <code>feature_id</code> , <code>feature_value_type</code> , and value fields (<code>float_value</code> , <code>int_array</code> , <code>int_value</code>).
<code>label</code>	<code>array[struct]</code>	Array of label dicts. Each element contains <code>action_time</code> and <code>action_type</code> .
<code>seq_feature</code>	<code>struct</code>	Sequence features dict with keys: <code>action_seq</code> , <code>content_seq</code> , <code>item_seq</code> . Each sub-key contains arrays of feature structs.
<code>timestamp</code>	<code>int64</code>	Event timestamp.
<code>user_feature</code>	<code>array[struct]</code>	Array of user feature dicts. Each element has <code>feature_id</code> , <code>feature_value_type</code> , and value fields (<code>float_array</code> , <code>int_array</code> , <code>int_value</code>).
<code>user_id</code>	<code>string</code>	User identifier.

Feature Struct Schema

Each feature element contains `feature_id`, `feature_value_type`, and several value fields. Depending on `feature_value_type`, the corresponding value fields are populated and the rest are `null`.

`item_feature` — value fields: `int_value`, `float_value`, `int_array`

```
{
  "feature_id": 6,
  "feature_value_type": "int_value",
  "float_value": null,
  "int_array": null,
```

```
"int_value": 96,  
}
```

user_feature — value fields: **int_value**, **float_array**, **int_array**

```
{  
  "feature_id": 65,  
  "feature_value_type": "int_value",  
  "float_array": null,  
  "int_array": null,  
  "int_value": 19  
}
```

seq_feature — value fields: **int_array**

```
{  
  "feature_id": 19,  
  "feature_value_type": "int_array",  
  "int_array": [1, 1, 1, ...]  
}
```

Possible **"feature_value_type"** values and their corresponding fields:

- **"int_value"** → **int_value**
- **"float_value"** → **float_value**
- **"int_array"** → **int_array**
- **"float_array"** → **float_array**
- Also there are some combinations of these types,
e.g. **"int_array_and_float_array"** → both **int_array** and **float_array** are populated.

Label Schema

Each element in the **label** array:

```
{  
  "action_time": 1770694299,  
}
```

```
"action_type": 1
}
```

Usage

```
import pandas as pd

df = pd.read_parquet("sample_data.parquet")
print(df.shape)          # (1000, 7)
print(df.columns)        # ['item_id', 'item_feature', 'label', 'seq_feature', 'timestamp', 'user_feature', 'user_id']
```

With Hugging Face `datasets`:

```
from datasets import load_dataset

ds = load_dataset("TAAC2026/data_sample_1000")
print(ds)
```

The dataset released in this competition is fully anonymized and does not reflect the exact production characteristics of Tencent's advertising platform.

Our dataset is a large-scale industrial dataset constructed from real-world advertising logs. It consists of two main components: (1) user behavior sequences and (2) non-sequential multi-field features.

User behavior sequences contain interaction events between users and items (e.g., exposure, click, conversion), each associated with side information such as timestamps and action types. Multi-field features include user attributes, item attributes, contextual signals, and cross features.

To ensure fairness and protect privacy, all sparse features are represented as anonymized integer IDs, and dense features are provided as fixed-length float vectors. No raw content (e.g., text, image, URL) or personally identifiable information is released.

In addition, we provide some example samples for reference:

The current samples are provided in JSON format, but the data used in the official competition may be adjusted based on this preliminary version, including possible changes to both the format and the actual content.

Sequential Data (e.g. one user behavior sequence)

```
{"user_id": "1", "seq": [{"item_id": 16612, "action_type": 1, "timestamp": 1770564000}, {"item_id": 49638, "action_type": 1, "timestamp": 1770564000}, ..., {"item_id": 173346, "action_type": 3, "timestamp": 1766960100}, ..., {"item_id": 49495, "action_type": 2, "timestamp": 1766576760}, ..., {"item_id": 1753, "action_type": 4, "timestamp": 1766399880}], ...}
```

User Features (e.g. one specific user)

```
[{"feature_id": 10, "feature_value_type": "int_array", "int_array": [1]}, // Marital Status

{"feature_id": 8, "feature_value_type": "int_value", "int_value": 1}, // Gender

{"feature_id": 7, "feature_value_type": "int_value", "int_value": 44}, ...] // Age
```

Item Features (e.g. one specific item)

```
[{"feature_id": 70, "feature_value_type": "int_value", "int_value": 2}, // Type

{"feature_id": 60, "feature_value_type": "int_value", "int_value": 3}, // Category

{"feature_id": 72, "feature_value_type": "int_value", "int_value": 2}, ...] // Advertiser Type
```

Context Features (e.g. one specific session)

```
[{"feature_id": 17, "feature_value_type": "int_value", "int_value": 3}, // Device Brand
```

```
{"feature_id": 21, "feature_value_type": "int_value", "int_value": 3}, ...] // OS Type
```

Cross Features

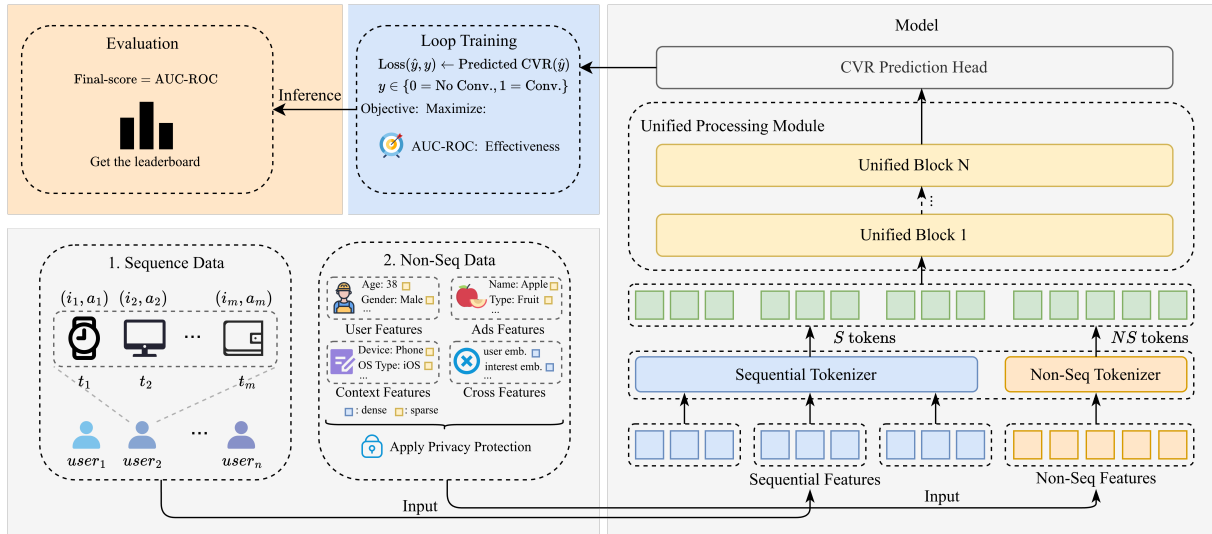
```
[{"feature_id": 25, "feature_value_type": "float_array", "float_array": [0.111, 0.057, 0.121, 0.043, -0.066, 0.081, 0.038, 0.105, -0.026, ...]}, ...] // User Embedding
```

Task

The main objective of this competition is to design a unified, stackable modeling block that simultaneously handles multi-field non-sequential tokens and sequential behavior tokens in one architecture.

Participants will be given the dataset described above, where each training/test instance corresponds to a triplet (user, context, target ad/item). The inputs consist of 1) Non-sequential multi-field features (user, ad, context, and cross features) and 2) A user behavior sequence with heterogeneous side information. The participants should build an effective model to capture the correlations between all these features, which outputs a predicted conversion rate (pCVR) for the target ad. We encourage (but not require) the participants to explore: 1) building stackable unified blocks over both sequence and non-sequence features (or tokens); 2) the scaling laws of the model. See Evaluation section for details. The overview of the competition framework is shown as Figure 1.

Figure 1. Overview of the competition framework.



Bottom-left (Input):

Each instance consists of (1) sequential behavior data, i.e., chronological user interaction histories with items and timestamps across multiple users, and (2) non-sequential multi-field features grouped into four categories: User Features, Ads Features, Context Features (sparse), and Cross Features (dense embeddings), all subject to privacy protection.

Right (A Demo Model):

The sequential and non-sequential features are converted into S tokens and NS tokens via dedicated tokenizers, then jointly processed by a stack of unified blocks within a single homogeneous backbone, followed by a CVR prediction head.

Top-left (Training & Evaluation):

The model is trained via loop-based optimization with a cross entropy loss. The final leaderboard score is the AUC of ROC score.

Evaluation

We will rank all teams using a single AUC of ROC metric (higher is better). To ensure practicality, each submission must also satisfy a track- and round-specific inference latency limit under the official evaluation environment and protocol; submissions that exceed the latency budget will be invalid and therefore not ranked, regardless of AUC.

To encourage innovation aligned with our theme—building a unified block that bridges sequence modeling and multi-field feature interaction, and exploring

the scaling law of recommendation—we will additionally offer two innovation awards: the Unified Block Innovation Award (\$45,000) and the Scaling Law Innovation Award (\$45,000). These awards are independent of leaderboard ranking. Final award decisions will be made by the committee based on a holistic review of the submitted technical reports, code, and the novelty and insights of the proposed methods, particularly along the two perspectives highlighted in this competition, rather than focusing solely on the final AUC score.

Awards

Prizes & Talent Opportunities

The challenge boasts a total prize pool of \$885,000, with dedicated tracks for academic and industrial teams, plus special innovation awards. Top performers also unlock exclusive Tencent career opportunities.

Unified Block Innovation Award \$45,000

This award will be decided by the organizing committee and honors the team that proposes and demonstrates the most compelling unified, stackable modeling block that effectively bridges sequential and non-sequential (multi-field) features.

Scaling Law Innovation Award \$45,000

This award will be decided by the organizing committee and honors the team that conducts the most rigorous and insightful exploration of scaling laws for recommendation models (e.g., how performance changes with #parameters, data, and compute scale).

References

1. [InterFormer: Effective Heterogeneous Interaction Learning for Click-Through Rate Prediction](#)
2. [OneTrans: Unified Feature Interaction and Sequence Modeling with One Transformer in Industrial Recommender](#)
3. [HyFormer: Revisiting the Roles of Sequence Modeling and Feature Interaction in CTR Prediction](#)